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 Studierichting: Informatica
 Oefeningenreeks 5A nr. 7.7

$$\begin{aligned}
 & \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} \\
 & t = \tan \frac{x}{2} \\
 & \sin x = \frac{2t}{1+t^2} \\
 & \frac{2}{1+t^2} dt = dx \\
 & \Rightarrow \int_0^1 \frac{1}{2 + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\
 & = \int_0^1 \frac{1}{\frac{2+2t^2}{1+t^2} + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\
 & = \int_0^1 \frac{2}{2t^2 + 2t + 2} dt \\
 & = \int_0^1 \frac{1}{t^2 + t + 1} dt \\
 & = \int_0^1 \frac{1}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}} dt \\
 & = \int_0^1 \frac{1}{\frac{3}{4} \left(\frac{\left(t + \frac{1}{2}\right)^2}{\frac{3}{4}} + 1 \right)} dt \\
 & = \frac{4}{3} \int_0^1 \frac{1}{\left(\frac{2\left(t + \frac{1}{2}\right)}{\sqrt{3}} \right)^2 + 1} dt \\
 & u = \frac{2t+1}{\sqrt{3}} \\
 & du = \frac{2}{\sqrt{3}} dt \\
 & \Rightarrow \frac{4\sqrt{3}}{6} \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{u^2 + 1} \\
 & = \frac{2\sqrt{3}}{3} [\text{Bgtan } u]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{2\sqrt{3}}{3} \left(\text{Bgtan}(\sqrt{3}) - \text{Bgtan}\left(\frac{1}{\sqrt{3}}\right) \right) \\
&= \frac{2\sqrt{3}}{3} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) \\
&= \frac{2\sqrt{3}}{3} \cdot \frac{\pi}{6} \\
&= \frac{2\sqrt{3}\pi}{18} = \frac{\sqrt{3}\pi}{9}
\end{aligned}$$

References